Matrix-Residual Graph Neural Networks for Graph Classification

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ABSTRACT

Background. Residual connections are widely used to stabilize graph neural networks, but residual design is often treated as a one-dimensional depth-wise choice. Multi-branch graph classifiers create a richer design space: residual information can flow along depth, across branches, or through a local two-dimensional branch-by-layer neighborhood.
Methods. We propose Matrix-Residual Graph Neural Networks as a controlled residual-topology study for graph classification. The model family compares five information-flow designs: plain message passing, vertical residual reuse, horizontal branch-wise exchange, local matrix residual reuse, and sparse/gated matrix reuse. The main benchmark evaluates six datasets (PROTEINS, DD, ENZYMES, MUTAG, AIDS, and Mutagenicity), four message-passing operators (GCNConv, GATConv, SAGEConv, and GINConv), five model families, and five graph-level folds. We further run branch-count ablations on PROTEINS and DD, fold-0 sensitivity scans for MatrixRes and MatrixResGated, five-fold checks of selected MatrixResGated candidates, and mechanism analyses based on branch diversity, branch cosine similarity, and gradient norms.
Results. No single residual topology dominates all datasets and operators. Across the 24 dataset–operator combinations, MatrixRes obtains the largest number of wins (9/24) and the best mean rank, followed by MatrixResGated. Under GCNConv, the winners differ by dataset: Plain is strongest on PROTEINS, HorizontalRes is strongest on DD, MatrixResGated is strongest on ENZYMES, and MatrixRes is strongest on MUTAG, AIDS, and Mutagenicity. Branch-count ablations reveal a non-monotonic rise-then-fall pattern: PROTEINS tolerates moderate branch expansion, whereas DD prefers smaller branch budgets. Five-fold tuned-candidate checks show that a smaller MatrixResGated hidden dimension reduces parameters on DD while retaining competitive accuracy; the PROTEINS sparsity candidate remains variable across folds.
Conclusions. Residual topology, branch count, and residual filtering should be selected jointly. Additional branches help only when they create useful functional diversity; otherwise, residual traffic and branch redundancy can offset the added capacity. Matrix-style reuse is therefore best understood as a tunable residual family rather than a universal replacement for simpler vertical or horizontal residual paths.

1 INTRODUCTION

Graph neural networks (GNNs) learn graph representations by repeatedly aggregating information from local neighborhoods (Gori et al., 2005; Kipf and Welling, 2017; Hamilton et al., 2017; Veličković et al., 2018; Xu et al., 2019). This message-passing view has become a standard tool for graph classification because it can combine node attributes, local neighborhoods, and higher-order graph structure in a single differentiable model (Gilmer et al., 2017; Duvenaud et al., 2015; Wu et al., 2021; Zhou et al., 2020). In molecular and protein-oriented benchmarks, the final prediction may depend on both local biochemical motifs and global graph organization, so the model must preserve useful intermediate evidence until graph-level pooling (Ying et al., 2018; Lee et al., 2019; Cangea et al., 2018). Increasing model depth or adding parallel branches can expand capacity, but it can also introduce over-smoothing, over-squashing,

redundant representations, unstable optimization, and unnecessary computation (Li et al., 2018; Oono and Suzuki, 2020; Chen et al., 2020b; Alon and Yahav, 2021; Topping et al., 2021; Zhao and Akoglu, 2020).

Residual connections are a common response to these difficulties. They improve gradient flow and make it possible to reuse earlier representations, a principle established in deep convolutional networks and later adapted to graph architectures (He et al., 2016; Bresson and Laurent, 2017; Li et al., 2019, 2021). Other depth-control strategies, including normalization, DropEdge, APPNP-style propagation, Jumping Knowledge aggregation, and graph-specific normalization, also aim to preserve useful information across propagation steps (Rong et al., 2019; Klicpera et al., 2018; Xu et al., 2018; Cai et al., 2021; Chen et al., 2020a). However, residual connectivity is still commonly treated as a default layer-wise component rather than as the object of study. Most residual graph classifiers reuse information along depth in the same stream. Once a model contains multiple branches, this view is incomplete: residual information may also move across branches at the same depth or through diagonal branch-depth neighborhoods.

Existing graph-classification work has explored powerful message-passing operators, hierarchical pooling, attention, and higher-order neighborhood mixing (Abu-El-Haija et al., 2019; Bianchi et al., 2020; Gao and Ji, 2019; Ying et al., 2018; Beaini et al., 2020; Dwivedi and Bresson, 2020). These designs improve the expressive or aggregation capacity of GNNs, but they do not isolate the residual-routing question: if several branch states are available at a given depth, should the model reuse only the previous state of the same branch, exchange states laterally, or combine both directions in a local two-dimensional residual neighborhood? Answering this question requires a controlled family in which the readout, optimizer, dataset split, and message-passing backbone are held fixed while only the residual topology changes.

This paper studies residual reuse as a structured design problem on a branch-by-layer grid. The central question is not whether skip connections should be present, but where residual signals should flow. A grid view exposes three controlled neighborhoods: vertical reuse along depth, horizontal exchange across neighboring branches, and matrix-style reuse that combines same-branch, neighboring-branch, and diagonal residual paths. A sparse/gated matrix variant further tests whether matrix residual traffic should be filtered before injection. This framing follows the spirit of controlled benchmark studies (Keriven et al., 2020; Morris et al., 2020): the goal is not to add a large number of unrelated architectural components, but to make a single design axis explicit enough to evaluate.

The study is deliberately controlled. It fixes the graph-classification pipeline within each comparison, evaluates the same architecture family under four standard message-passing operators, and changes only the residual topology and residual-control mechanism. This separation is important because it keeps the empirical comparison focused on information-flow design rather than on unrelated changes in readout structure or task-specific feature engineering.

The work is organized around four research questions:

1. **Which residual topology is strongest under a shared graph-classification protocol?** We compare plain message passing, vertical residual reuse, horizontal branch-wise exchange, local matrix residual reuse, and sparse/gated matrix reuse on six graph-classification datasets under four message-passing operators.

2. **How does branch count change residual behavior?** We test $B = 1, \ldots, 8$ for the multi-branch residual families and ask whether more branches improve accuracy monotonically or only within a useful operating region.

3. **Why include sparse/gated matrix reuse?** We test whether matrix residual traffic should be passed densely or filtered before injection.

4. **When does controlled matrix reuse help?** We use sensitivity scans, five-fold tuned-candidate checks, and mechanism metrics to distinguish promising residual-control settings from fold-specific artifacts.

The main contributions are:

- **A branch-by-layer residual formulation for graph classification.** Vertical, horizontal, matrix, and sparse/gated matrix residual families are expressed as local neighborhoods on the same grid, making their differences explicit while preserving a common backbone and classifier.

- **A controlled five-fold benchmark and branch-count ablation.** The experiments cover 24 dataset–operator combinations. MatrixRes obtains the largest number of wins, but the per-dataset winners remain mixed, and branch-count curves follow a rise-then-fall pattern.

- **Evidence that branch expansion is not merely a capacity knob.** Branch diversity, branch cosine similarity, and gradient norms show that additional branches help only when they create useful separation without excessive residual traffic or optimization weakening.

- **A conservative validation of MatrixResGated tuning.** Fold-0 sensitivity scans identify candidate operating regions, and five-fold reruns show that the smaller DD MatrixResGated setting is parameter-efficient, whereas the PROTEINS sparsity candidate is not yet a robust full-fold gain.

2 MATERIALS AND METHODS

This section defines the graph-classification task, the Matrix-Residual architecture family, the benchmark protocol, and the mechanism metrics. The guiding principle is to isolate residual topology while holding the remaining learning pipeline fixed.

2.1 Problem definition and notation

Let $\mathscr{G} = (\mathscr{V}, \mathscr{E})$ be a graph with node feature matrix $\mathbf{X} \in \mathbb{R}^{n \times d_{\text{in}}}$ and adjacency structure $\mathbf{A}$. For a dataset $\mathscr{D} = \{(\mathscr{G}_i, y_i)\}_{i=1}^{N}$, the objective is to learn $f_\theta : \mathscr{G} \to \hat{y}$, where $\hat{y}$ is the predicted graph label.

The compared model families share the same graph-classification wrapper:

- **Plain**: stacked message passing without explicit residual reuse.

- **VerticalRes**: same-branch residual reuse from the previous layer.

- **HorizontalRes**: cross-branch exchange between adjacent branches at the same depth.

- **MatrixRes**: local two-dimensional reuse over vertical, horizontal, and diagonal neighbors.

- **MatrixResGated**: matrix reuse with gated or sparse residual filtering.

2.2 Shared graph-classification pipeline

Let B be the number of branches and L be the number of message-passing layers. The hidden state at branch b and layer ℓ is written as $\mathbf{H}_b^{(\ell)}$, where $b \in \{1, \ldots, B\}$. A branch-local propagation step is

$$\mathbf{Z}_b^{(\ell)} = \phi_b^{(\ell)} \left(\mathbf{H}_b^{(\ell-1)}, \mathbf{A} \right), \tag{1}$$

where ϕ is instantiated as one of four standard operators: GCNConv (Kipf and Welling, 2017), GATConv (Veličković et al., 2018), SAGEConv (Hamilton et al., 2017), or GINConv (Xu et al., 2019). The residual-enhanced update is

$$\mathbf{H}_b^{(\ell)} = \mathbf{Z}_b^{(\ell)} + \sum_{(b',\ell') \in \mathscr{N}(b,\ell)} \mathscr{R}_{(b',\ell') \to (b,\ell)} \left(\mathbf{H}_{b'}^{(\ell')} \right). \tag{2}$$

The only term that changes across architectures is the residual neighborhood $\mathscr{N}(b,\ell)$ and the residual transform $\mathscr{R}$.

After the final message-passing layer, each branch is pooled to a graph-level vector by global mean pooling and the branch vectors are combined before classification. We intentionally avoid architecture-specific pooling heads such as differentiable pooling or self-attention pooling (Ying et al., 2018; Lee et al., 2019), because the purpose is to isolate residual topology rather than to compare readout mechanisms. The classifier is trained with cross-entropy loss.

2.3 Residual neighborhoods

Plain. The plain baseline has no residual neighborhood:

$$\mathcal{N}(b,\ell) = \emptyset. \tag{3}$$

VerticalRes. Vertical residual reuse injects the previous layer on the same branch:

$$\mathcal{N}(b,\ell) = \{(b,\ell-1)\}. \tag{4}$$

HorizontalRes. Horizontal residual reuse exchanges information across adjacent branches at the same depth:

$$\mathcal{N}(b,\ell) = \{(b-1,\ell),(b+1,\ell)\}, \tag{5}$$

with boundary branches using only valid neighbors.

MatrixRes. Matrix residual reuse combines vertical, horizontal, and diagonal neighborhoods:

$$\mathcal{N}(b,\ell) = \{(b,\ell-1),(b-1,\ell),(b+1,\ell),(b-1,\ell-1),(b+1,\ell-1)\}. \tag{6}$$

This local stencil is intentionally not dense all-to-all mixing. It preserves an interpretable residual structure over the branch-by-layer grid.

MatrixResGated. MatrixResGated uses the same neighborhood as MatrixRes, but the residual transform controls the injected signal:

$$\mathscr{R}(\mathbf{U}) = \begin{cases} \alpha\mathbf{U}, & \text{identity residual filtering with a gate,} \\ \alpha\,\mathrm{softshrink}_{\lambda}(\mathbf{U}), & \text{sparse residual filtering with a gate.} \end{cases} \tag{7}$$

In the main benchmark, Plain, VerticalRes, HorizontalRes, and MatrixRes use identity residual filtering. MatrixResGated uses sparse residual filtering with $\lambda = 0.05$ and a learnable scalar gate initialized at 0.8. The implementation also supports top-k residual filtering and fixed gates, but these are not the default settings used for the main reported results.

2.4 Datasets and evaluation protocol

The main benchmark uses six datasets from TUDataset (Morris et al., 2020): PROTEINS, DD, ENZYMES, MUTAG, AIDS, and Mutagenicity. PROTEINS and DD are binary protein-structure graph classification benchmarks, ENZYMES is a six-class enzyme classification benchmark, and MUTAG, AIDS, and Mutagenicity provide supplementary molecular-graph robustness checks. All datasets are loaded through PyTorch Geometric (Fey and Lenssen, 2019). All experiments use the raw benchmark features and graph structures without task-specific feature engineering, graph augmentation, or edge redesign.

All main results use graph-level five-fold evaluation. For fold k, the test accuracy is

$$\mathrm{Acc}_k = \frac{1}{|\mathscr{T}_k|} \sum_{(\mathscr{G}_i,y_i)\in\mathscr{T}_k} \mathbf{1}[\hat{y}_i = y_i]. \tag{8}$$

Across folds, we report

$$\overline{\mathrm{Acc}} = \frac{1}{5}\sum_{k=1}^{5} \mathrm{Acc}_k, \tag{9}$$

and the fold-level standard deviation.

2.5 Training configuration

All main benchmark runs use the default branch count $B = 3$. The benchmark is repeated under four message-passing operators: GCNConv, GATConv, SAGEConv, and GINConv. The hidden dimension is 64 unless a sensitivity setting changes it, and all model families use the same optimizer, early-stopping logic, graph pooling, and fold protocol. Models are optimized with Adam (Kingma and Ba, 2014); the checkpoint with the lowest validation loss is retained for test evaluation. The branch-count ablation evaluates $B = 1,\ldots,8$ for HorizontalRes, MatrixRes, and MatrixResGated on PROTEINS and DD under GCNConv. The sensitivity scan evaluates MatrixRes and MatrixResGated on fold 0 at $B = 3$, varying learning rate, dropout, hidden dimension, sparsity strength, and gate initialization. Selected MatrixResGated candidates are then rerun across all five folds.

Table 1. Key experimental settings shared by the main benchmark.

Parameter	Value
Message-passing operators	GCNConv, GATConv, SAGEConv, GINConv
Default branch count	3
Default hidden dimension	64
Evaluation	Five-fold graph classification
Branch-count ablation	$B = 1, \ldots, 8$
Primary metric	Mean best test accuracy
Sensitivity scope	Fold 0, $B = 3$
Early stopping	Validation loss, patience 60 or 80

2.6 Mechanism metrics

To interpret branch-count behavior, we track three mechanism indicators. Branch diversity measures the mean pairwise distance between branch representations. Branch cosine similarity measures whether branches remain redundant. Mean gradient norm tracks optimization weakening as branch count and residual traffic grow. CKA and residual-traffic summaries are retained as supplementary diagnostics.

3 RESULTS

All results are reported as five-fold mean best test accuracy $\pm$ fold-level standard deviation unless otherwise stated.

3.1 Overall benchmark performance

The main benchmark asks whether a single residual topology remains strongest across datasets and message-passing operators. The answer is negative. Across the 24 dataset–operator combinations, MatrixRes obtains the largest number of wins (9/24) and the best mean rank (2.13), but it is not uniformly dominant. MatrixResGated wins 5 combinations, Plain wins 5, HorizontalRes wins 3, and VerticalRes wins 2. This mixed ordering is important for the paper's central claim: matrix-style reuse is a strong residual topology, but it should be treated as a tunable design family rather than as a universal replacement for simpler residual paths.

Table 2 reports the GCNConv slice at the default branch count $B = 3$. This slice is useful because it holds the operator fixed while showing how the preferred topology changes by dataset. Plain is strongest on PROTEINS, HorizontalRes is strongest on DD, MatrixResGated is strongest on ENZYMES, and MatrixRes is strongest on MUTAG, AIDS, and Mutagenicity.

Table 2. GCNConv benchmark results with default branch count $B = 3$.

Dataset	Plain	VerticalRes	HorizontalRes	MatrixRes	MatrixResGated
PROTEINS	$\mathbf{0.7044 \pm 0.0387}$	0.6993 ± 0.0340	0.6992 ± 0.0399	0.6973 ± 0.0324	0.6886 ± 0.0397
DD	0.7053 ± 0.0517	0.7143 ± 0.0260	$\mathbf{0.7181 \pm 0.0296}$	0.7127 ± 0.0365	0.7141 ± 0.0359
ENZYMES	0.2683 ± 0.0343	0.2550 ± 0.0455	0.2633 ± 0.0584	0.2750 ± 0.0577	$\mathbf{0.2933 \pm 0.0470}$
MUTAG	0.7556 ± 0.0439	0.7451 ± 0.0695	0.7343 ± 0.0433	$\mathbf{0.7609 \pm 0.0715}$	0.7556 ± 0.0721
AIDS	0.8215 ± 0.0046	0.8210 ± 0.0233	0.8130 ± 0.0297	$\mathbf{0.8365 \pm 0.0412}$	0.8150 ± 0.0231
Mutagenicity	0.7644 ± 0.0188	0.7667 ± 0.0212	0.7738 ± 0.0134	$\mathbf{0.7784 \pm 0.0185}$	0.7701 ± 0.0165

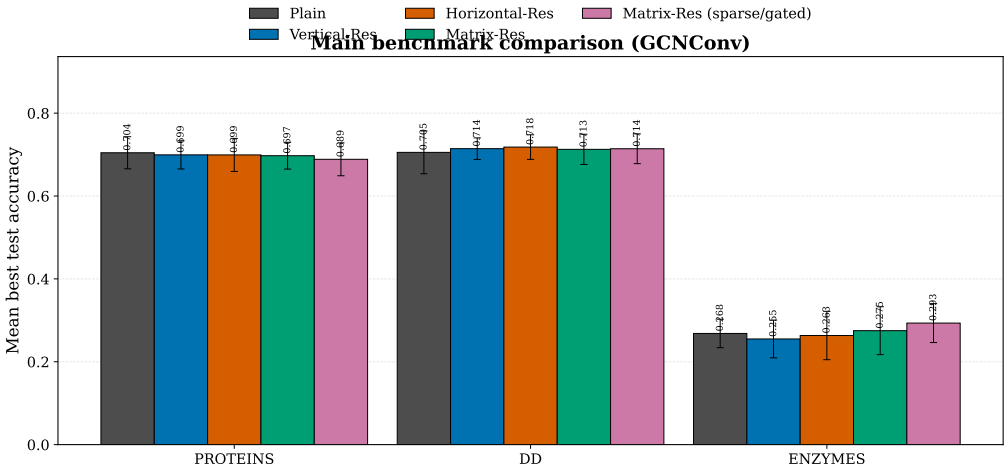

Figure 1. Main GCNConv benchmark at $B = 3$. Bars show five-fold mean best test accuracy with fold-level standard-deviation error bars.

Figure 1 makes the split regime visually explicit. The default GCNConv setting does not support a universal residual winner; instead, each dataset favors a different residual neighborhood.

3.2 Operator robustness

The full four-operator benchmark tests whether the GCNConv pattern is operator-specific. Table 3 counts the winning topology across the 24 dataset–operator combinations. MatrixRes is the most frequent winner overall and wins under every operator family, but the remaining wins are distributed across all other models. This confirms that the residual-topology question is not reducible to one backbone choice.

Table 3. Winner counts across 24 dataset–operator combinations.

Model	Wins	Mean rank
Plain	5	3.04
VerticalRes	2	3.38
HorizontalRes	3	3.54
MatrixRes	**9**	**2.13**
MatrixResGated	5	2.92

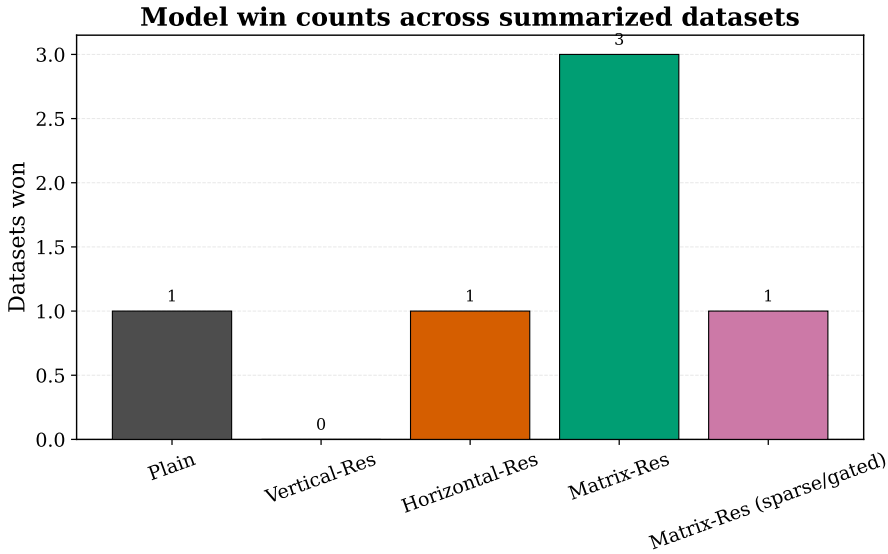

Figure 2. Model-level winner counts across the six datasets and four message-passing operators.

3.3 Branch-count ablation

Branch count has a non-monotonic effect. Table 4 reports the best branch count for each topology in the ablation. On PROTEINS, HorizontalRes peaks at $B = 4$, MatrixRes peaks at $B = 6$, and MatrixResGated peaks at $B = 3$. On DD, the preferred branch budget is smaller: HorizontalRes and MatrixRes peak at $B = 2$, while MatrixResGated is strongest at $B = 1$.

Table 4. Best branch-count setting for each topology in the ablation.

Dataset	Model	Best B	Accuracy
PROTEINS	HorizontalRes	4	0.7125 ± 0.0286
PROTEINS	MatrixRes	6	0.7062 ± 0.0210
PROTEINS	MatrixResGated	3	0.7098 ± 0.0219
DD	HorizontalRes	2	0.7275 ± 0.0304
DD	MatrixRes	2	0.7206 ± 0.0411
DD	MatrixResGated	1	0.7283 ± 0.0244

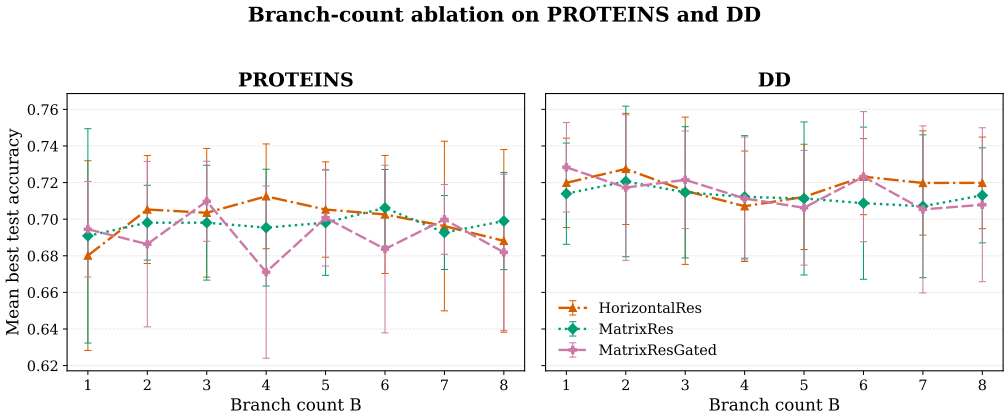

Figure 3. Branch-count ablation for HorizontalRes, MatrixRes, and MatrixResGated on PROTEINS and DD. Accuracy improves only within a useful operating region and then saturates or declines.

The ablation shows that branch count should not be read as a simple capacity knob. Increasing B

changes the residual graph itself: it adds branches, residual paths, residual traffic, and optimization interactions. PROTEINS tolerates a moderate branch budget, while DD is more sensitive to redundant branch expansion.

3.4 MatrixResGated sensitivity

The fold-0 sensitivity scan shows that MatrixResGated is tunable, but its preferred region differs between datasets. On PROTEINS, the baseline MatrixResGated configuration reaches 0.6726, and mild sparsification with $\lambda = 0.02$ improves fold-0 accuracy to 0.6951. Stronger sparsification with $\lambda = 0.1$ reduces accuracy to 0.6278. On DD, the baseline fold-0 score is 0.7300, while $lr = 0.001$, $dim = 32$, and $gate_init = 0.2$ each reach 0.7511.

Table 5. Strongest MatrixResGated fold-0 sensitivity settings.

Dataset	Setting	Accuracy	Interpretation
PROTEINS	baseline	0.6726	default controlled matrix reuse
PROTEINS	$\lambda = 0.02$	0.6951	mild sparsification improves fold 0
PROTEINS	$\lambda = 0.1$	0.6278	strong sparsification is harmful
DD	baseline	0.7300	default controlled matrix reuse
DD	$lr = 0.001$	0.7511	conservative optimization helps
DD	$dim = 32$	0.7511	smaller hidden dimension helps
DD	$gate_init = 0.2$	0.7511	lower initial gate helps

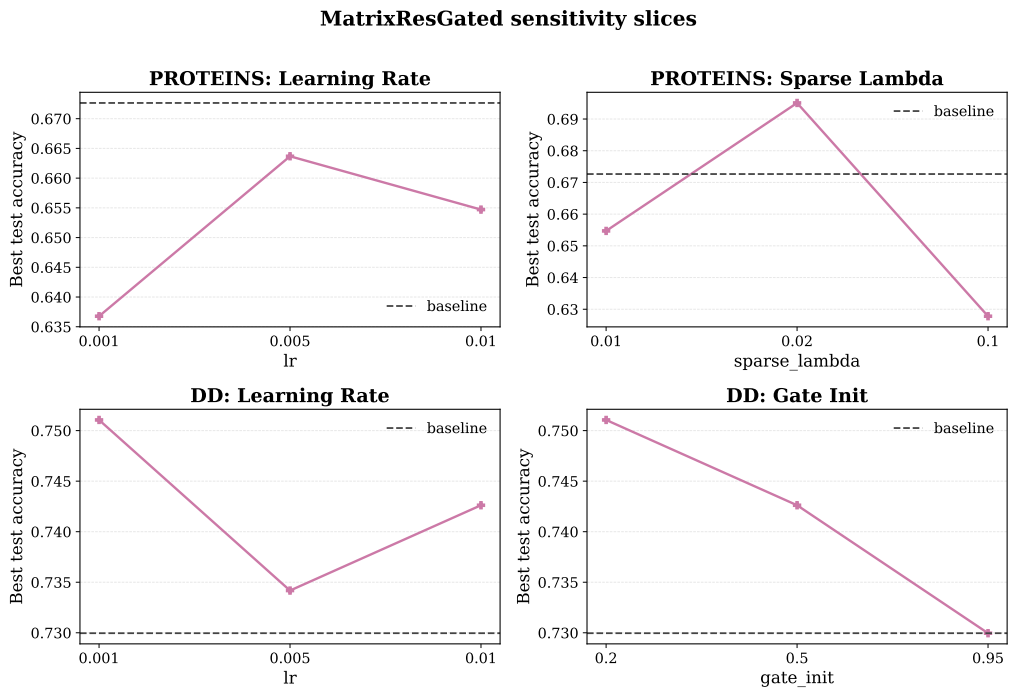

Figure 4. Fold-0 MatrixResGated sensitivity scan at $B = 3$. The scan identifies candidate operating regions but is not treated as final hyperparameter evidence.

3.5 Five-fold check of tuned MatrixResGated candidates

To distinguish candidate discovery from evaluation, selected MatrixResGated settings were rerun across all five folds. Table 6 reports these follow-up checks. On DD, the smaller hidden dimension is the strongest tuned candidate, reaching 0.7215 ± 0.0330 while reducing parameters from 42,371 to 15,043. The lower learning rate is essentially tied with the default MatrixResGated benchmark, and the lower gate initialization is weaker. On PROTEINS, $\lambda = 0.02$ reaches 0.6909 ± 0.0447, below the default

MatrixResGated result, showing that the fold-0 sparsity gain is not stable across all folds in the current run.

Table 6. Five-fold checks of selected MatrixResGated candidates.

Candidate	Dataset	Accuracy	Mean loss	Parameters
`PROTEINS_sparse_lambda_0.02`	PROTEINS	0.6909 ± 0.0447	0.6104	38,339
`DD_lr_0.001`	DD	0.7181 ± 0.0336	0.5737	42,371
`DD_dim_32`	DD	$\mathbf{0.7215 \pm 0.0330}$	0.5717	15,043
`DD_gate_init_0.2`	DD	0.7131 ± 0.0213	0.5820	42,371

3.6 Mechanism analysis

The mechanism summaries support a common explanation for the rise-then-fall trend. Increasing branch count initially creates useful diversity: branches become less identical and can cover complementary feature subspaces. Past the useful regime, diversity alone is not sufficient. Residual traffic grows quickly, branch similarity may remain high, and gradient norms can weaken.

For HorizontalRes on PROTEINS, moving from $B = 1$ to $B = 4$ increases branch diversity from 0.0000 to 1.1450, while branch cosine decreases from 1.0000 to 0.8417. This corresponds to the accuracy improvement. Beyond $B = 4$, diversity continues increasing, reaching 1.5441 at $B = 8$, but accuracy declines to 0.6881. MatrixRes shows a similar trend at a larger branch budget, peaking at $B = 6$. DD is more conservative: branch expansion beyond $B = 2$ often adds residual complexity without enough functional separation.

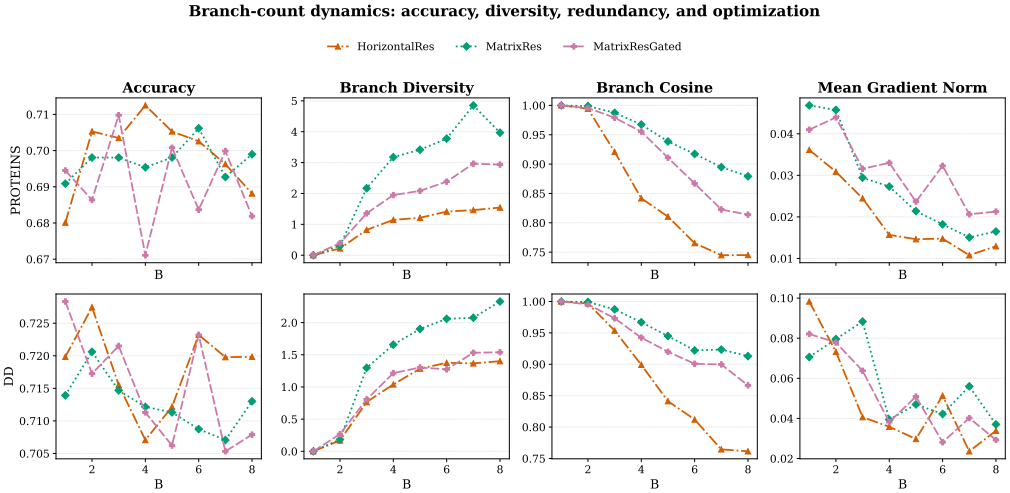

Figure 5. Mechanism summary linking branch-count accuracy to branch diversity, branch cosine similarity, and mean gradient norm.

4 DISCUSSION

This section answers the research questions in light of the six-dataset, four-operator benchmark, the branch-count ablation, the tuned-candidate checks, and the mechanism evidence.

4.1 RQ1: Which residual topology is strongest?

The main benchmark argues against a universal residual topology. Across 24 dataset–operator combinations, MatrixRes is the most frequent winner and has the best mean rank, but all five model families win at least two combinations. Under GCNConv alone, Plain is strongest on PROTEINS, HorizontalRes is strongest on DD, MatrixResGated is strongest on ENZYMES, and MatrixRes is strongest on MUTAG, AIDS, and Mutagenicity. These outcomes indicate that the useful residual neighborhood depends on dataset-specific tolerance for branch specialization, residual traffic, and optimization complexity.

This conclusion is consistent with the broader GNN literature: changing the propagation operator, normalization, or pooling design often shifts the preferred architecture (Xu et al., 2019; Hamilton et al., 2017; Veličković et al., 2018; Ying et al., 2018). The present study adds that residual topology itself is another source of this shift. Matrix-style reuse can be read as a local residual stencil over the branch-by-layer grid, analogous in spirit to structured neighborhood mixing (Abu-El-Haija et al., 2019), but applied to hidden-state reuse rather than to graph adjacency. Its advantage is therefore architectural organization, not a new message-passing primitive.

Answer: MatrixRes is the strongest overall topology in the current full benchmark, but residual topology is still dataset- and operator-dependent. A graph-classification benchmark should compare vertical, horizontal, matrix-style, and sparse/gated reuse rather than assuming that one residual direction is universally best.

4.2 RQ2: How does branch count change residual behavior?

The branch-count ablation shows that B is not only a capacity parameter. Increasing B adds branches, residual paths, and optimization interactions. On PROTEINS, moderate expansion is useful: HorizontalRes peaks at $B = 4$, MatrixRes at $B = 6$, and MatrixResGated at $B = 3$. On DD, smaller budgets are preferable, with peaks at $B = 1$ or $B = 2$. The mechanism summaries explain this split. Branch diversity may increase with B, but if branch cosine remains high or gradients weaken, the extra branches do not translate into better classification.

This rise-then-fall behavior is also a practical warning for residual GNN design. Residual connections are often introduced to make deeper or wider models easier to optimize (He et al., 2016; Li et al., 2019, 2021), but a multi-branch residual model changes both capacity and routing density. A larger B increases the number of paths through which hidden states can be reinjected, so the useful regime is governed by the balance between functional diversity and residual interference. The branch-count results suggest that this balance differs sharply between PROTEINS and DD, even though both are commonly treated as protein-structure graph benchmarks.

Answer: Branch count helps when it creates useful functional diversity, and hurts when residual complexity grows faster than branch separation or optimization quality.

4.3 RQ3: Why include sparse/gated matrix reuse?

MatrixResGated tests whether matrix residual traffic should be filtered before injection. The main benchmark shows that sparse/gated reuse is not uniformly better than dense matrix reuse: MatrixRes wins more dataset–operator combinations overall. However, MatrixResGated is still useful because it wins several combinations, including the difficult ENZYMES GCNConv setting, and it provides a controlled way to reduce weak residual coefficients. This distinction is important. The value of MatrixResGated is not that sparsity always improves accuracy, but that sparse residual control can be beneficial when matrix-style reuse would otherwise inject too much low-value residual traffic.

This places sparse/gated residual reuse between two familiar design ideas. Like residual gated graph networks, it learns how strongly historical information should be injected (Bresson and Laurent, 2017); like sparsification and regularization methods, it reduces the influence of weak or noisy pathways (Rong et al., 2019; Cangea et al., 2018). The present implementation is deliberately simple: it uses a scalar gate and soft-shrink filtering rather than a large routing network. This keeps the ablation interpretable and prevents the gated variant from becoming a separate high-capacity model.

Answer: Sparse/gated matrix reuse is a control mechanism for residual traffic. It is complementary to dense MatrixRes rather than a strict replacement.

4.4 RQ4: When does controlled matrix reuse help?

MatrixResGated defines a tunable family rather than a plug-and-play winner. The fold-0 sensitivity scan suggests that PROTEINS can benefit from mild sparsification, while DD benefits from conservative optimization, smaller hidden dimension, and lower initial gate. The five-fold candidate reruns refine this interpretation. On DD, the smaller hidden dimension remains beneficial and parameter-efficient. On PROTEINS, the mild-sparsity candidate does not preserve its fold-0 advantage across all folds.

Answer: Controlled matrix reuse is useful, but candidate settings must be validated across folds. The current evidence supports a smaller controlled matrix model on DD and leaves PROTEINS sparsification as a promising but not yet stable direction.

4.5 Limitations

Several limitations bound the generality of these findings. First, although the main benchmark includes four message-passing operators, it still uses a fixed graph-level pooling and classification head; alternative readout designs may interact with residual topology. Second, the sensitivity scan is fold-0 only; although selected candidates were rerun across five folds, that follow-up is limited to MatrixResGated on PROTEINS and DD. Third, ENZYMES has low absolute accuracy and high variability, so its result should be interpreted cautiously. Finally, the mechanism analysis is correlational. Branch diversity, cosine similarity, and gradient norms support a coherent explanation, but they do not prove causality.

Another limitation is that the benchmark intentionally avoids task-specific chemical or biological feature engineering. This makes the residual-topology comparison cleaner, but it also means that the absolute accuracies should not be interpreted as optimized state-of-the-art molecular prediction results. The paper's contribution is methodological: it identifies when different residual pathways are helpful under matched conditions.

4.6 Future directions

Future work should extend the comparison to larger graph-classification datasets, richer readout mechanisms, and adaptive residual-neighborhood selection. A natural next step is to replace manually chosen vertical, horizontal, or matrix stencils with a learned local residual policy. The tuned-candidate validation should also be broadened beyond the current MatrixResGated checks to determine whether dataset-specific residual control can be selected reliably.

5 CONCLUSIONS

This paper studies residual connectivity for graph classification as a structured branch-by-layer design problem. Across six datasets and four message-passing operators, residual topology matters, but its value depends on the dataset, operator, and branch budget. MatrixRes is the strongest overall topology by winner count and mean rank, but the per-dataset GCNConv winners remain mixed: Plain is strongest on PROTEINS, HorizontalRes on DD, MatrixResGated on ENZYMES, and MatrixRes on MUTAG, AIDS, and Mutagenicity. The branch-count ablations further show that adding branches is useful only within a limited operating region.

The mechanism analysis explains why branch expansion is not automatically beneficial. Additional branches can increase representational diversity, but they also increase residual traffic and may preserve redundant branch similarity or weaken optimization. Matrix-style reuse is therefore best viewed as a controlled residual family whose effectiveness depends on the branch budget and residual filtering strategy.

The practical implication is that residual topology, branch count, operator choice, and residual control should be evaluated jointly. Before treating a sensitivity result as a performance claim, candidate settings should be rerun across folds. In the current experiments, this validation supports a smaller and more parameter-efficient MatrixResGated model on DD, while the PROTEINS sparsity candidate remains less stable.

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A APPENDIX

A.1 Result sources

The manuscript uses the following summary files as the source of truth. The main benchmark summary contains 120 model-level rows: six datasets, four operators, five model families, and five-fold aggregation for each combination.

- `records/LATEST/summaries/benchmark_summary.csv`
- `records/LATEST/summaries/branch_ablation_summary.csv`
- `records/LATEST/summaries/parameter_sensitivity_summary.csv`

- `records/LATEST/summaries/tuned_candidate_summary.csv`
- `records/LATEST/summaries/mechanism_compact_summary.csv`

A.2 Supplementary diagnostics

Additional fold-level rows, CKA summaries, residual traffic summaries, and full sensitivity tables are retained in the repository records directory for supplementary conversion.

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